



NEXT CHAPTER @ SMITH PUBLIC LIBRARY

Shapes, Space & the Calculator

Session 6 — Geometry, Volume & the On-Screen Calculator

Monday Evenings • 5:30–7:30 PM • Library Conference Room

Tonight's Agenda

1

Welcome & quick recap

10 min

2

Perimeter & area

25 min

3

Volume of a box

20 min

4

Cylinders & cones

25 min

5

The on-screen calculator

20 min

6

Worked real-world example

10 min

7

Wrap-up & take-home preview

10 min

Quick Recap: Last Week

Last week:

we pictured equations as lines. Slope tells you how steep, and the intercept tells you where it starts.

Tonight:

we measure real shapes: the fence around a yard, the floor of a room, and how much a can or a box can hold.

Perimeter vs. Area



Perimeter = Around

The distance around the outside of a shape. Think: a fence. Measured in plain units, like feet.



Area = Inside

The flat space inside a shape. Think: flooring. Measured in square units, like square feet.

Rectangles: Perimeter & Area



length = 9

width = 4

Perimeter: add all four sides.

$$P = 2 \times \text{length} + 2 \times \text{width}$$

Area: length times width.

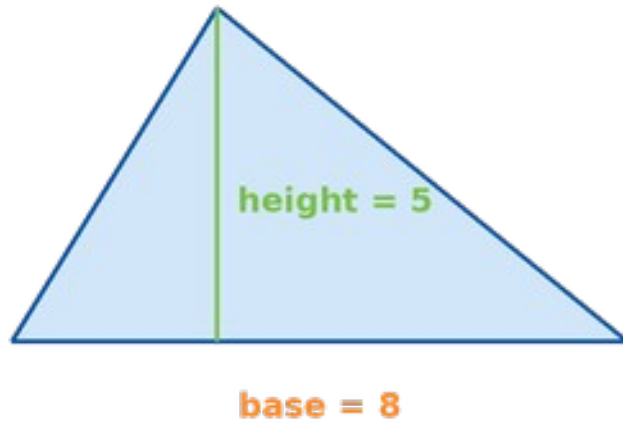
$$A = \text{length} \times \text{width}$$

Example: 9 by 4

$$P = 2(9) + 2(4) = 18 + 8 = 26 \text{ units}$$

$$A = 9 \times 4 = 36 \text{ square units}$$

Triangles: Area



A triangle is half of a rectangle.

$$A = \frac{1}{2} \times \text{base} \times \text{height}$$

Use the straight-up height, not the slanted side.

Example: base 8, height 5

$$A = \frac{1}{2} \times 8 \times 5 = 20 \text{ square units}$$

Your Turn: Perimeter & Area

GUIDED PRACTICE

1

A rectangle is 12 units long and 5 units wide. What is its perimeter?

2

A rectangle is 7 feet long and 6 feet wide. What is its area?

3

A triangle has a base of 10 and a height of 6. What is its area?

4

Fencing costs \$3 per foot. A yard is 20 feet by 10 feet. How much will it cost to fence all the way around it?

Let's Check Your Answers



1. $2(12) + 2(5) = 24 + 10 = 34$ units



2. $7 \times 6 = 42$ square feet

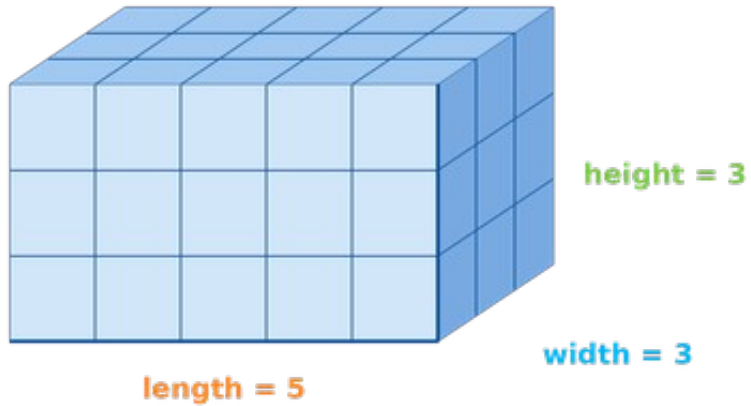


3. $\frac{1}{2} \times 10 \times 6 = 30$ square units



4. Perimeter = $2(20) + 2(10) = 60$ feet, so $60 \times \$3 = \180

Volume of a Box



Volume is how much a 3-D shape holds.

Count the layers: each layer is length $\times$ width, then stack them up by the height.

$$V = \text{length} \times \text{width} \times \text{height}$$

Example: 5 by 3 by 3

$$V = 5 \times 3 \times 3 = 45 \text{ cubic units}$$

Units are cubed: cubic inches, cubic feet.

Your Turn: Volume of a Box

GUIDED PRACTICE

1

A box is 4 inches long, 3 inches wide, and 2 inches tall. What is its volume?

2

A storage bin is 10 inches long, 6 inches wide, and 5 inches tall. What is its volume?

3

A fish tank is 20 inches long, 10 inches wide, and 12 inches tall. How many cubic inches of water can it hold?

4

A cube has sides of 3 inches. What is its volume?

Let's Check Your Answers



1. $4 \times 3 \times 2 = 24$ cubic inches



2. $10 \times 6 \times 5 = 300$ cubic inches



3. $20 \times 10 \times 12 = 2,400$ cubic inches



4. $3 \times 3 \times 3 = 27$ cubic inches (a cube has equal length, width, and height)

Three Volume Formulas



Box

$$V = l \times w \times h$$

Layers of length $\times$ width, stacked to the height.



Cylinder

$$V = \pi \times r^2 \times h$$

The circle on the bottom ($\pi \times r^2$) times the height.



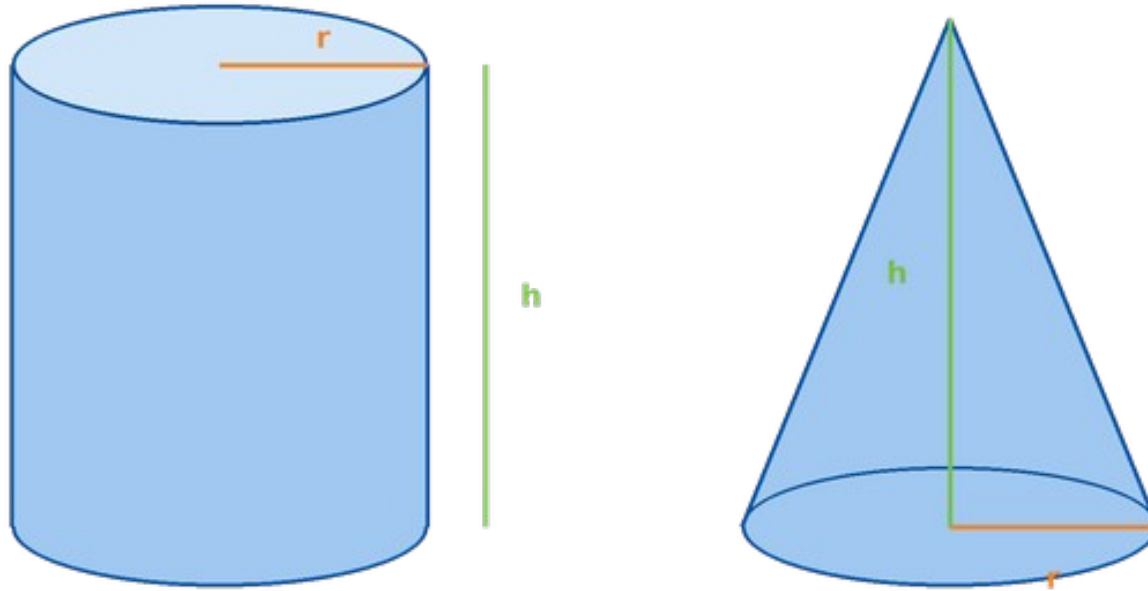
Cone

$$V = \frac{1}{3} \times \pi \times r^2 \times h$$

One-third of a cylinder with the same base and height.

The GED puts these on the formula sheet for you. Your job: pick the right one and use it.

Cylinder and Cone: Same Base, Same Height



r = radius

distance from the center of the circle to its edge.

h = height

straight up, not slanted.

Watch for diameter!

The radius is half of the diameter.

A cone holds exactly $\frac{1}{3}$ as much as the cylinder around it.



Cylinder, Step by Step

A can has a radius of 4 inches and a height of 3 inches. Find its volume.

Write the formula: $V = \pi \times r^2 \times h$

Square the radius: $4^2 = 16$

Multiply: $16 \times 3 = 48$

So the volume is 48π cubic inches.

With the calculator: $\pi \times 48 \approx 150.8$ cubic inches

For a cone, do the same and then divide by 3.

Cone with $r = 3$, $h = 6$: $3^2 \times 6 = 54$, $\div 3 = 18$, so $18\pi \approx 56.5$ cubic inches.

Your Turn: Cylinders & Cones

GUIDED PRACTICE

1

A cylinder has a radius of 3 inches and a height of 4 inches. What is its volume, to the nearest tenth?

2

A cone has a radius of 6 inches and a height of 2 inches. What is its volume, to the nearest tenth?

3

A can is 10 inches across the top (diameter) and 8 inches tall. What is its volume, to the nearest tenth?

4

A cylinder holds 90 cubic inches. A cone has the same base and height. How much does the cone hold?

Let's Check Your Answers



1. $\pi \times 3^2 \times 4 = 36\pi \approx 113.1$ cubic inches



2. $\frac{1}{3} \times \pi \times 6^2 \times 2 = 24\pi \approx 75.4$ cubic inches



3. Radius = $10 \div 2 = 5$, so $\pi \times 5^2 \times 8 = 200\pi \approx 628.3$ cubic inches



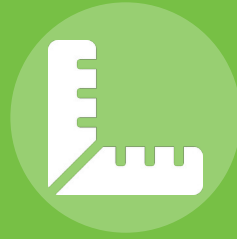
4. $90 \div 3 = 30$ cubic inches (a cone is $\frac{1}{3}$ of the cylinder)

Using the On-Screen Calculator



Use the π key

Press π instead of typing 3.14. It is faster and more exact.



Square with x^2

Type the radius, press x^2 , then multiply by the rest.



Round at the end

Keep every digit on the screen. Round only after the last step.

On test day: about the first 5 math questions have no calculator. After that, the on-screen calculator opens.



Calculator Walk-Through

Cylinder with radius 4 and height 3

Press: $\pi \times 4 \times^2 \times 3$ enter

Screen: 150.7964474

Round to the nearest tenth: 150.8

Cone with radius 3 and height 6

Press: $\pi \times 3 \times^2 \times 6 \div 3$ enter

Screen: 56.54866776

Round to the nearest tenth: 56.5

Let's Solve One Together

A can is 6 inches across the top and 10 inches tall. How much does it hold? Round to the nearest tenth.

Watch the trap:

“across the top” is the diameter, 6. The radius is half, so $r = 3$.

Formula: $V = \pi \times r^2 \times h$

Plug in: $V = \pi \times 3^2 \times 10 = \pi \times 9 \times 10 = 90\pi$

Calculator: $\pi \times 90 \approx 282.7$ cubic inches

Before You Go



Try tonight's take-home practice

Real practice questions, plus an Extra Practice tab with a Challenge Round.



Thursday office hours are open

Bring a shape or formula that feels shaky and we'll work through it.



Next week: Math Checkpoint

The official GED Ready: Math practice test, then a debrief on your results.



You Can Measure Anything.

See you next Monday for the Math Checkpoint.

Before next week:

Finish tonight's take-home practice and look over your cheat sheets from Sessions 2 through 6.
GED.com's free test preview tutorials (tools.ged.com) are always open, in English or Spanish.